

Griseofulvin Loaded Microemulsion Gel for Topical Antifungal Therapy

- **Task 2:** Case Study – Drug Formulation.
- **Dosage Form:** Topical Microemulsion Gel.
- **Drug:** Griseofulvin.
- Advanced formulation approach to improve solubility, skin penetration and local antifungal activity.

Introduction

Need for Formulation ??

- Griseofulvin is a lipophilic antifungal drug (BCS Class II)
- Used for dermatophyte infections
- Problems: poor aqueous solubility and variable oral bioavailability
- Microemulsion gel improves topical delivery and reduces systemic exposure

Formulation Process

1. Solubility screening of drug in oils, surfactants and co-surfactants
2. Selection of optimized microemulsion region using pseudo-ternary phase diagram
3. Preparation of oil + surfactant/co-surfactant mixture
4. Addition of aqueous phase
5. Incorporation into gel base (Carbopol/HPMC)

Excipients Used and Their Functions

- **Oil phase:** Oleic acid / Isopropyl myristate – improves drug solubility
- **Surfactant:** Tween 80 – reduces interfacial tension
- **Co-surfactant:** Ethanol / Propylene glycol – enhances stability & penetration
- **Gelling agent:** Carbopol / HPMC – provides viscosity and controlled release

Evaluation and Stability Factors

Evaluation parameters:

- Droplet size & PDI
- Zeta potential
- pH and viscosity
- Drug content
- In-vitro release and skin permeation

Stability concerns: phase separation, drug precipitation, viscosity changes

Challenges and Conclusion

Challenges:

- High surfactant concentration may cause irritation
- Maintaining viscosity and release balance
- Scale-up difficulties

Conclusion:

- Griseofulvin microemulsion gel enhances solubility, penetration, controlled release and patient compliance.